

Faculty of Electrical Engineering

WARSAW UNIVERSITY OF TECHNOLOGY

1DA1636:A Automotive Electrical Systems

PROJECT

TITLE

SIGNAL CONDITIONING CIRCUIT FOR INDUCTIVE SPEED SENSOR(RPM METER)

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1. Introduction

The objective of the project is to create a working signal conditioning circuit for a speed sensor which will let us observe the 0V, 5V square wave signal at the output, which is mainly feedable into microcontrollers without causing a damage to them. Project has been implement on a breadboard, using basic hardware components, meaning no software is to be expected from this projects implementation. The sensor signal is a sine wave with some noise, with amplitude ranging from 6V to 24V peak-to-peak, centered near 0V. The goal of the signal conditioning circuit is to convert this variable-amplitude sine wave into a stable 0V–5V square wave suitable for direct microcontroller input without risk of damage.

Initially, IC-based approaches using a Schmitt trigger (SH74HC14N) and a comparator with hysteresis (LM393P) were explored. Both proved unsuccessful in practice due to input voltage range incompatibility and biasing issues. The final working solution uses a diode bridge rectifier cascaded with two BJT stages to achieve the required output.

2. Literature review

Literature was consulted mainly to support the IC-based approaches tried in the early stages of the project. The SN74HC14N datasheet [1] was used to understand the input voltage limits and how to properly connect the Schmitt trigger. The LM393P datasheet [2] was used for the comparator circuit with hysteresis, following the reference design from the TI application guide [4]. For understanding the full-wave diode bridge rectification and how it handles AC signals with both positive and negative half-cycles, basic electronics resources were consulted as this is a well known technique for this kind of application. Both ICs were ultimately unsuccessful. Following the professor's recommendation to use a discrete transistor approach, Razavi's 'Fundamentals of Microelectronics' [3] was consulted for common collector and common emitter biasing configurations, which form the basis of the final circuit visible in Figure 3.

[1] Documentation SH74HC14N
[2] Documentation LM393P
[3] ‘Fundamentals of microelectronics’ Razavi
[4] Guide by TI for comparator hysteresis circuit

3. Methodology

The development process began with measuring the output voltage waveforms of the inductive sensor, which ranged from 6V to 24V peak-to-peak centered near 0V, as shown in Figures 4 and 5. Figure 1 shows the complete laboratory measurement setup. Figure 2 shows the inductive sensor itself.
First I want to list the hardware components used. These were used in final successful solution:

4 Diodes
2 Bjts - BC547C
5 Resistors
5V power supply

Also I want to note about my failed attempts hardware components:
Comparator - LM393P
Schmiddt - SH74HC14N
Zener diodes

The Schmitt trigger circuit (SH74HC14N) is shown in Figure 6, and the comparator circuit (LM393P) is shown in Figure 7. The Schmitt trigger failed because its input voltage is limited to the supply voltage range of 5V, and the sensor signal with peaks up to 12V exceeded the maximum allowable input voltage. The comparator circuit was attempted with a zener diode clamp, but reliable switching thresholds could not be achieved across the full 6V–24V input amplitude range. Resistive voltage dividers were used during testing to verify threshold levels, but the output remained unstable. These failures showed that IC-based solutions required significantly more supporting circuitry than anticipated, making the discrete transistor approach more practical.

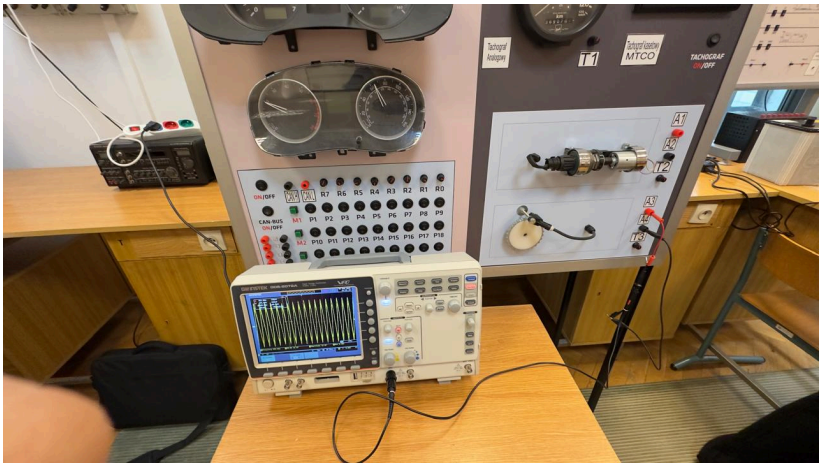


Figure 1 - sensor output measurement set-up

Figure 2 is to show the inductive sensor specifically to not be confused with other devices in figure 1.

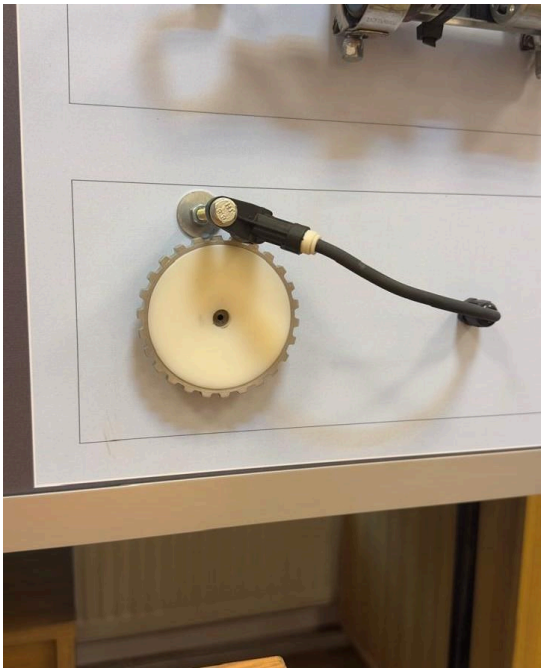


Figure 2 - inductive sensor

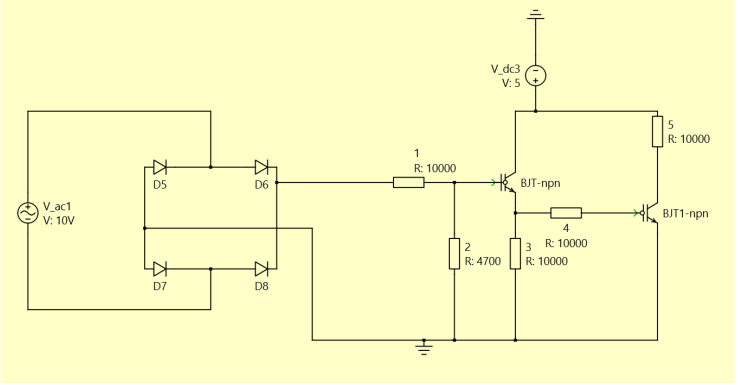


Figure 3 - final circuit schematic

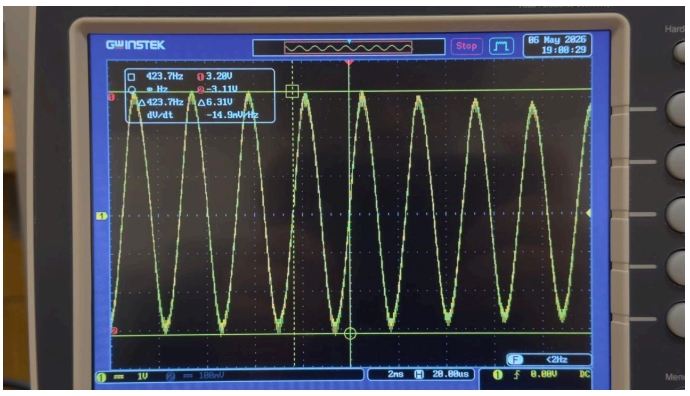


Figure 4 - minimal voltage output



Figure 5 - maximal voltage output

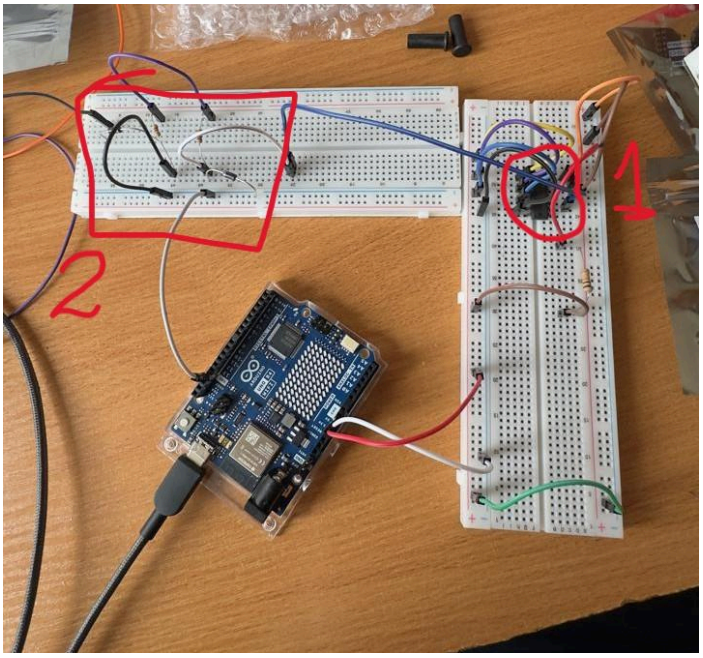


Figure 6 - schmitt trigger circuit. 1 - schmitt trigger IC, 2 - diode with zener and resistors,to rectify and voltage regulator.

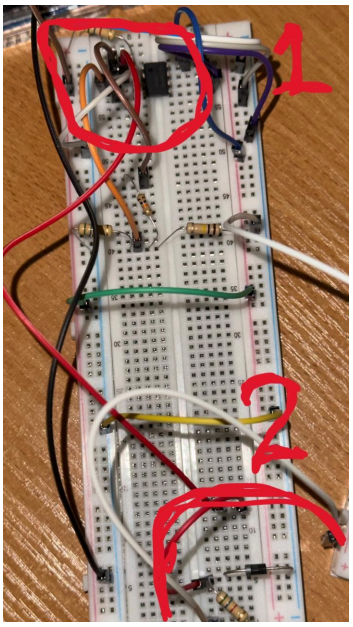


Figure 7 - comparator circuit. 1 - comparator IC, 2 - zener with resistor.

Unsuccessful circuits took a decent time to complete, research the components buy the ones suitable and read their documentations and learn how to properly connect them. And not being able to check themwith the input cost some time. My checking method was using resistive voltage dividers and checking if the thresholds are working correctly.

The correct and final solution is shown in Figure 8 as a breadboard implementation and in Figure 3 as a circuit schematic. The circuit operates in two stages. The first stage is a full-wave diode bridge (D5–D8) which rectifies the AC input, converting both positive and negative half-cycles into positive pulses. The rectified signal passes through resistor R1 (10kΩ) to the base of the first BJT (BC547C), configured as a common collector stage. Resistors R2 (4.7kΩ) and R3 (10kΩ) set the operating point. The collector is connected to 5V. The output is taken from the emitter and fed through R4 (10kΩ) to the base of the second BJT (BC547C), configured as a common emitter stage. Resistor R5 (10kΩ) is the collector load. The output is taken from the collector of this second stage, producing a clean 0V–5V square wave.

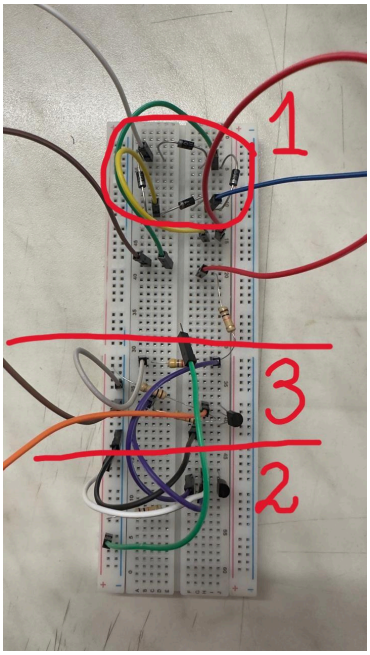


Figure 8 - solution with cascaded transistors

4. Results and Discussion

The circuit is successful. The output voltage waveform is shown in Figure 9, confirming a clean 0V–5V square wave. Figure 10 confirms the output amplitude is 5V as expected. Figure 11 shows the frequency measurement. The output frequency is twice that of the input signal due to the full-wave rectifier bridge — both the positive and negative half-cycles of the input are converted to positive pulses, doubling the pulse rate. For practical RPM calculation, the measured pulse frequency must therefore be divided by two. The output is suitable for direct microcontroller input. The circuit was verified at both the minimum (6V peak-to-peak, Figure 4) and maximum (24V peak-to-peak, Figure 5) sensor output levels, and produced a stable 5V output in both cases.

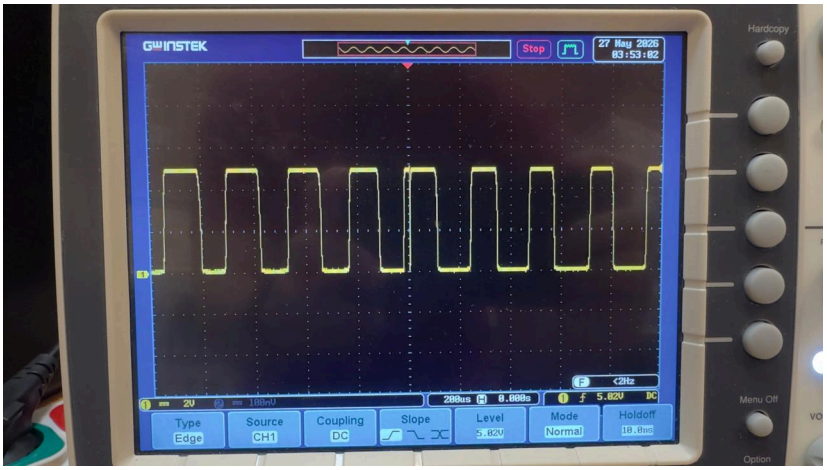


Figure 9 - the output waveform(0-5V)

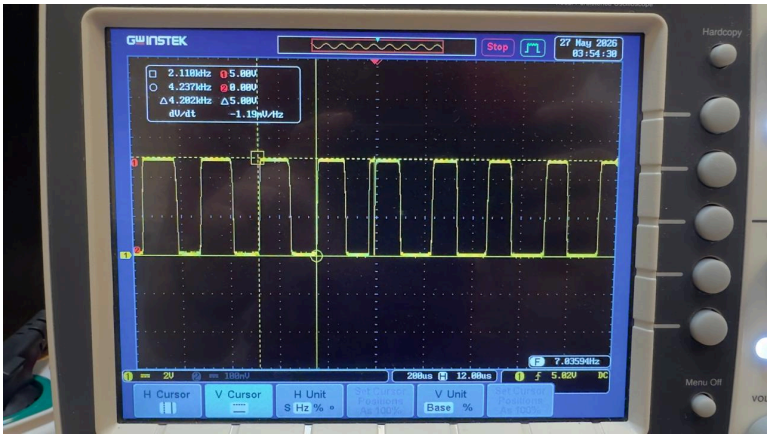


Figure 10 - Amplitude measurement

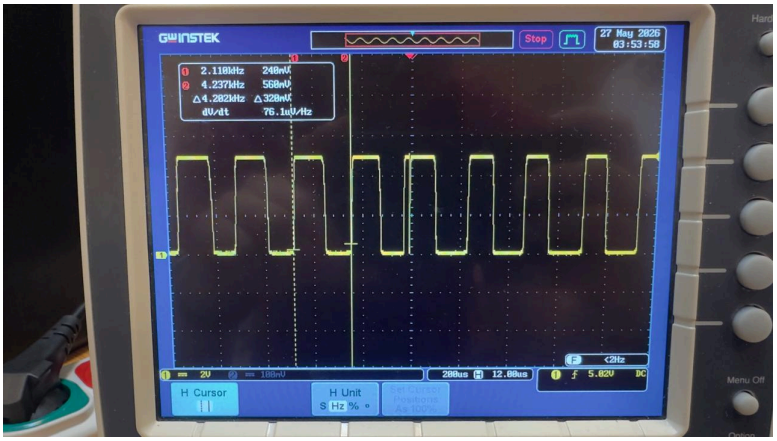


Figure 11 - Frequency measurement

5. Conclusions

All project objectives were successfully achieved. The final circuit, shown in Figure 8, produces a stable 0V–5V square wave output across the full sensor input range of 6V–24V peak-to-peak, with output frequency proportional to rotational speed.

The main difficulty was the time spent on two IC-based approaches (Figures 6 and 7) that did not work. Each required component research, sourcing, and breadboard implementation. The inability to test with the actual sensor signal during development significantly prolonged debugging.

If this project were extended, a natural next step would be adding a frequency-to-voltage converter, allowing RPM to be read directly as an analog voltage by a microcontroller ADC. A PCB implementation would also improve noise immunity compared to the breadboard prototype shown in Figure 8.

References

[1] Documentation SH74HC14N -
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